

Learning Resource On Speech and Video Processing

Module-2: Feature Extraction and Speech Recognition

Chapter-6: Dynamic Features and Normalization, Vector Quantization

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Chapter Outcomes

- After completing this module, students will be able to:
 - Explain dynamic features, including Delta and Delta-Delta coefficients, in speech processing.
 - Calculate Delta and Delta-Delta coefficients from given speech feature vectors using basic equations.
 - Explain the need for feature normalization
 - Describe common normalization techniques used in speech processing.
 - Explain the basic concept of Vector Quantization and its role in representing speech feature vectors.
 - Apply a basic Vector Quantization procedure to quantize speech feature vectors using distance-based matching.

Static vs Dynamic Features

- In **speech processing**, features extracted from each short-time speech frame can be broadly classified as **static** features and **dynamic** features.
- **Static features** describe the characteristics of speech at a particular frame, whereas the **dynamic features** describe how speech characteristics change across frames.
- While the **static features** represent current spectral/phonetic characteristics, the **dynamic features** represent rate and acceleration of feature changes.
- **Static features** are useful in capturing the current state of speech. On the other hand, **dynamic features** captures speech transitions and temporal patterns.
- **Static features:** MFCC, LPCC, Spectral features
- **Dynamic features:** Delta, Delta-Delta coefficients

Why dynamic features???

- Speech is **not a stationary** signal. Its characteristics continuously change with time as we move from one sound to another.
- Therefore, using only static features from individual frames may not capture the complete information present in speech.
- **Dynamic features** are needed due to following reasons:
 - Speech changes its properties over time.
 - Static features describe only the current frame.
 - Speech transitions from one sound to another sound contain important information.
 - Dynamic features capture temporal information like rate of change (Delta), and change in that rate (Delta-Delta).

Delta (Δ) Coefficients

- **Delta (Δ) Coefficients:** A Delta coefficient represents the rate of change of a speech feature over time. Mathematically, it can be represented as below:

$$\Delta c_t = \frac{c_{t+1} - c_{t-1}}{2}$$

where,

c_t is the feature value at the current frame.

c_{t+1} is the feature value at the next frame.

c_{t-1} is the feature value at the previous frame.

Δc_t is the rate of change of the feature.

Sample Example

Suppose the three consecutive MFCCs are 10, 12, and 16 then determine the delta coefficients.

Sol:

Given,

$$c_{t-1} = 10; c_{t+1} = 16, \text{ and } c_t = 12$$

So,

$$\Delta c_t = \frac{c_{t+1} - c_{t-1}}{2} = (16 - 10) / 2 = 3$$

- The positive value of delta (Δ) indicates that the feature is **increasing with time.**

Delta-Delta ($\Delta\Delta$) Coefficients

- – **Delta-Delta ($\Delta\Delta$) Coefficients:** A Delta-Delta coefficient represents the rate of change of the Delta coefficient.
 - It is therefore often described as the **acceleration** of the speech feature.
 - Mathematically, delta-delta coefficients can be represented as below:

$$\Delta\Delta\mathbf{c}_t = \frac{\Delta\mathbf{c}_{t+1} - \Delta\mathbf{c}_{t-1}}{2}$$

where, $\Delta\mathbf{c}_{t-1}$ is delta coefficient of the previous frame;
 $\Delta\mathbf{c}_{t+1}$ is delta coefficient of the next frame; $\Delta\Delta\mathbf{c}_t$ is the delta-delta coefficient of the current frame.

Sample Example

Consider the following Delta values for three consecutive frames: $\Delta c_{t-1} = 2$, $\Delta c_t = 3$, and $\Delta c_{t+1} = 5$ then determine delta-delta coefficient.

Sol:

Given,

$$\Delta c_{t-1} = 2, \Delta c_t = 3, \text{ and } \Delta c_{t+1} = 5$$

So,

$$\Delta \Delta c_t = \frac{\Delta c_{t+1} - \Delta c_{t-1}}{2} = 5 - 2 / 2 = 1.5$$

- The positive value of delta-delta ($\Delta \Delta$) indicates that the rate of change is itself **increasing with time**.

Example

Consider five consecutive frames of one MFCC coefficient:

$$c_{t-2} = 8; c_{t-1} = 10; c_t = 12; c_{t+1} = 16; c_{t+2} = 18$$

Determine the delta and delta-delta coefficients.

Sol:

Given,

$$c_{t-1} = 10; c_{t+1} = 16, \text{ and } c_t = 12$$

So,

$$\Delta c_t = \frac{c_{t+1} - c_{t-1}}{2} = (16 - 10) / 2 = 3$$

Now, we need to calculate

$$\Delta \Delta c_t = \frac{\Delta c_{t+1} - \Delta c_{t-1}}{2}$$

So, To calculate the Delta at frame $t - 1$, we use the frames immediately before and after $t - 1$ i.e., $t - 2$ and t .

$$\Delta c_{t-1} = \frac{c_t - c_{t-2}}{2} = \frac{12 - 8}{2} = 2$$

i.e., the rate of change of the MFCC around frame $t - 1$ is 2.

Now, To calculate the Delta at frame $t + 1$, we use the frames immediately before and after $t + 1$ i.e., t and $t + 2$.

$$\Delta c_{t+1} = \frac{c_{t+2} - c_t}{2} = \frac{18 - 12}{2} = 3$$

i.e., the rate of change of the MFCC around frame $t + 1$ is 3.

Now, the delta-delta coefficient can be calculated as below:

$$\Delta \Delta c_t = \frac{\Delta c_{t+1} - \Delta c_{t-1}}{2} = \frac{3 - 2}{2} = 0.5$$

- The positive Delta-Delta value (+0.5) indicates that the rate of

Need for Normalization

- **Normalization** is the process of transforming speech feature values so that they have a **similar and consistent** scale or distribution.
 - In speech processing, features such as MFCC, LPCC, and Delta coefficients can have different numerical ranges.
 - Normalization reduces these variations and makes the features more suitable for machine-learning or speech-recognition systems.
- **Need for normalization**
 - Different speakers may produce different feature values.
 - Recording device and conditions can change feature values.
 - Speaking volume and intensity can introduce variations.
 - Features may have different numerical scales, causing¹¹ some features to have more influence than others.

Mean and Variance Normalization

- **Mean Normalization:**

- First, calculate the **mean** of the feature values

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i;$$

where, μ is mean; N is the number of frames; x_i is the feature value at i-th frame; $i=0,1,2,\dots,N$

- Then, **subtract** the mean value from each feature value.

$$x'_i = x_i - \mu$$

- **Variance Normalization:** To make the spread of the values consistent, divide x'_i by the standard deviation.

$$x''_i = \frac{x'_i}{\sigma}$$

Sample Example

Suppose the values of one MFCC coefficient from four speech frames are: 8, 10, 12, 14.

Sol:

S-1: Calculate the mean

$$\mu = \frac{8 + 10 + 12 + 14}{4} = 11$$

S-2: Subtract the mean $x_i - \mu$

$$x_1 - \mu = 8 - 11 = -3$$

$$x_2 - \mu = 10 - 11 = -1$$

$$x_3 - \mu = 12 - 11 = 1$$

$$x_4 - \mu = 14 - 11 = 3$$

Mean Normalization is complete.

S-3: Calculate standard deviation

$$\sigma^2 = \frac{(-3)^2 + (-1)^2 + (1)^2 + (3)^2}{4} = 5$$

$$\text{Therefore, } \sigma = \sqrt{5} = 2.236$$

A small standard deviation means the feature values are close to their mean, while a large standard deviation means they are more widely spread.

S-4: Divide by standard deviation

$$x'_1 = \frac{8 - 11}{2.236} = -1.34$$

Similarly, the other normalized values are calculated.

Hence, the normalized values are:

Cepstral Mean Normalization

- **Cepstral Mean Normalization (CMN)** is a specific application of mean normalization to cepstral features, such as MFCCs.
- Calculate the mean of each cepstral coefficient over a speech segment and subtract that mean from each coefficient.

- For coefficient k : $\mathbf{c}'_{t,k} = \mathbf{c}_{t,k} - \mu_k$;

where,

$\mathbf{c}_{t,k}$ is k th cepstral coefficient at frame t ;

μ_k is mean of the k th coefficient over the speech segment;

$\mathbf{c}'_{t,k}$ is normalized coefficient

CMN vs Mean Normalization

- CMN is mainly used to reduce relatively constant effects introduced by the recording environment or channel, such as microphone and transmission characteristics.
- Mean normalization is a general feature-normalization technique, whereas CMN is mean normalization specifically applied to cepstral features such as MFCCs.
- Mean normalization makes the feature mean approximately zero, whereas CMN reduces channel/recording-condition effects in cepstral features

Vector Quantization

- **Vector Quantization (VQ)** is used to represent a large number of speech feature vectors using a small set of representative vectors called **codewords**.
- When speech is processed, each frame can produce a feature vector such as an MFCC vector: $X_t = [c_1, c_2, c_3, \dots, c_n]$
- Working with these large number of vectors needs more computation storage as well as time. It can be addressed by VQ as below:
 - Reducing the number of vectors that results in less computational storage and time.
 - VQ reduces the computational complexity.
 - Similar MFCC vectors can be grouped and represented

Vector Quantization Terminologies

- **Vector:** Collection of feature values representing one speech frame.
 - For example, if we extract 3 MFCC coefficients from a speech frame: $X=[4.2, 7.5, 3.6]$. This is a 3-dimensional feature vector.
- **Codeword:** A codeword is a representative vector that represents a group of similar feature vectors.
 - Suppose several MFCC vectors are: $[2.0, 3.4, 1.9]$, $[2.2, 3.6, 1.7]$, $[2.1, 3.5, 1.8]$. These vectors are quite similar. VQ can represent them using one representative vector: $C_1 = [2.1, 3.5, 1.8]$
- **Codebook:** A codebook is a collection of all the codewords.
 - $C = \{C_1, C_2, C_3, C_4\}$;

Distance measures (Euclidean distance)

- In Vector Quantization (VQ), we need to determine which codeword is most similar to a given speech feature vector.
 - To do this, we calculate the distance between the input vector and each codeword. The codeword with the smallest distance is considered the closest match.
- **Euclidean Distance** measures the straight-line distance between two vectors.
- For two n-dimensional vectors $X=[x_1, x_2, x_3, \dots, x_n]$ and $C=[c_1, c_2, c_3, \dots, c_n]$, the **Euclidean Distance** is

$$d(\mathbf{x}, \mathbf{c}) = \sqrt{\sum_{i=1}^n (\mathbf{x}_i - \mathbf{c}_i)^2}$$

where, $d(\mathbf{x}, \mathbf{c})$ is the Euclidean Distance; x_i is the i th vector; c_i

Vector Quantization Encoding Process

- The basic VQ encoding process is as below:
 - Collect the speech data
 - Extract the Feature Vectors
 - Group the similar vectors and create codeword accordingly.
 - Obtain the Codebook
 - Calculate the Euclidean distance between the input vector and each codeword.
 - Select the Nearest Codeword

Sample Example

Suppose an input MFCC vector is: $x = [2, 3]$ and we have two codewords: $C1 = [1, 2]$ and $C2 = [5, 4]$. Determine the given vector x belongs to which codeword.

Sol:

S-1: E.D of the given vector x with the codewords $C1$ and $C2$

$$d(x, c1) = \sqrt{(2 - 1)^2 + (3 - 2)^2} = 1.41$$

$$d(x, c2) = \sqrt{(2 - 5)^2 + (3 - 4)^2} = 3.16$$

The vector always belongs to the closest codeword i.e., to the codeword with the lower E.D.

Hence, the given vector x belongs to the codeword $C1$.

Codebook Generation

- **Codebook generation** is the process of creating a set of representative vectors (codewords) from a large collection of training speech feature vectors.
- The basic process of codebook generation is as below:
 - Collect training vectors
 - Initialize codewords (randomly selected)
 - Assign vectors to the nearest codeword
 - For each group, calculate the mean/centroid of all the vectors assigned to that group. The centroid becomes the new codeword.
 - The assignment and update steps are repeated.
 - The final representative vectors form the codebook.

Application of VQ in Speech Processing

- **Vector Quantization** can be used in speech processing in following applications:
 - **Speaker Recognition:** VQ can be used to create a speaker-specific codebook.
 - **Speech Coding:** VQ can be used to compress speech information.
 - **Speech Recognition:** VQ can provide a compact representation of speech features.
 - **Speech Storage:** VQ provides a compact representation, it can reduce the storage required for speech-related feature information.